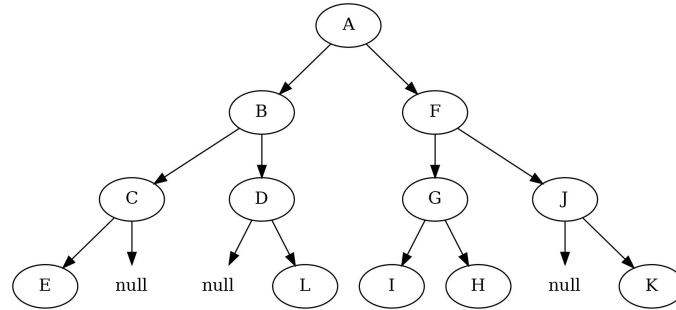


Algorithms and Data Structures

29.06.2023

Exercise 1 (1.5 points)

The following binary tree is given.



Visit it in post-order. Please, report the sequence of keys displayed by the visit on the same line and separate letters with a single space. No other symbols must be included in the response. This is an example of the response format: B D E A etc.

Solution

Tree visit

Pre-order:	A	B	C	E	D	L	F	G	I	H	J	K
In-order:	E	C	B	D	L	A	I	G	H	F	J	K
Post-order:	E	C	L	D	B	I	H	G	K	J	F	A

Response

E C L D B I H G K J F A

Exercise 2 (1.5 points)

A BST contains integer values included in the range 1-1000. Suppose we are looking for the value 531 in such a BST. Consider the following sequences of values generated during a search.

```
259 681 524 653 600 590 539 535 533 526 530 531
323 935 693 376 676 560 428 313 544 531
907 306 577 470 495 560 541 539 508 533 515 529 532 531
776 475 680 617 502 612 505 526 539 536 433 528 530 531
```

Indicate which ones of the following sequences (numbered 1, 2, 3, and 4) are correct. Notice that wrong responses imply a mark penalty.

Solution

BST verify sequence

Searching: 531

Sequence 1: 259 681 524 653 600 590 539 535 533 526 530 531

259 681 524 653 600 590 539 535 533 526 530 531 Sequence OK.

Sequence 2: 323 935 693 376 676 560 428 313 544 531

323 Error: 313<323

Sequence 3: 907 306 577 470 495 560 541 539 508 533 515 529 532 531

907 306 577 470 495 560 541 539 508 533 515 529 532 531 Sequence OK.

Sequence 4: 776 475 680 617 502 612 505 526 539 536 433 528 530 531

776 475 Error: 433<475

Response

1 3

Exercise 3 (1.5 points)

Suppose we have an initially empty priority queue implemented with a maximum heap. Consider the following sequence of integers and "*" characters, where each integer corresponds to one insertion into the priority queue and each character "*" corresponds to one extraction.

10 2 4 12 5 3 * *

Report the sequence of values as they are stored in the array representing the priority queue at the end of the entire process. Please, show the entire content of the array as a sequence of integer values separated by a single space. No other symbols must be included in the response. This is an example of the response: 0 3 2 6 8 etc.

Solution

```
### Heap insertion and extraction
Max-Heap.
Insert keys: 10 2 4 12 5 3 * *
Insert 10: 10
Insert 2: 10 2
Insert 4: 10 2 4
Insert 12: 12 10 4 2
Insert 5: 12 10 4 2 5
Insert 3: 12 10 4 2 5 3
Extract 12: 10 5 4 2 3
Extract 10: 5 3 4 2
```

Response

5 3 4 2

Exercise 4 (1.5 points)

In an activity set, the i -th activity is identified by the pair $[s_i, f_i)$, where s_i is the starting time and f_i is the finishing time. The activities are numbered starting from 1. The following is a correct set of activities.

```
P1 18 21
P2 5 9
P3 23 26
P4 4 7
P5 2 4
P6 27 31
P7 29 33
P8 28 30
P9 14 20
P10 30 34
```

Using a greedy algorithm, find the largest subset of mutually compatible activities. Please, report the set of compatible activities in the same order they have been selected, separated by a single space. No other symbols must be included in the response. This is an example of the response: 1 3 2 6 8

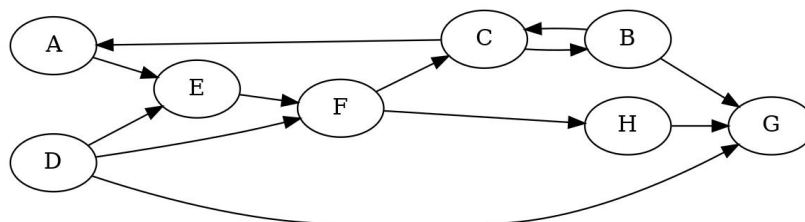
Solution

```
### Activity selection
Number of activities: 10
P1 18 21
P2 5 9
P3 23 26
P4 4 7
P5 2 4
P6 27 31
P7 29 33
P8 28 30
P9 14 20
P10 30 34
Selected activities:
P5 [ 2, 4)
P4 [ 4, 7)
P9 [ 14, 20)
P3 [ 23, 26)
P8 [ 28, 30)
P10 [ 30, 34)
```

Response
5 4 9 3 8 10

Exercise 5 (1.5 points)

Given the following directed graph, represent the reverse graph and find all strongly connected components.



Compute all strongly connected components in the graph. Report them in alphabetic order and within each component indicates vertices in alphabetic order (i.e., ACZ and not CAZ or ZAC). Separate components with a single space. No other symbols must be included in the response. This is an example of the response format: ABC EF XY etc.

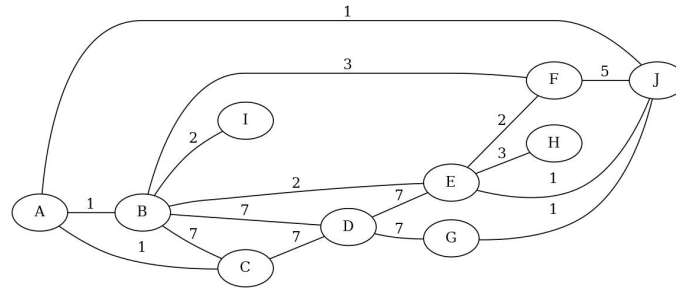
Solution

```
Directed matrix ... stated as such.
UN-weighted matrix ...stated as such.
Vertex list: [ 0] A [ 1] B [ 2] C [ 3] D [ 4] E [ 5] F [ 6] G [ 7] H
Adjacency Matrix:
0 0 0 0 1 0 0 0
0 0 1 0 0 0 1 0
1 1 0 0 0 0 0 0
0 0 0 0 1 1 1 0
0 0 0 0 0 1 0 0
0 0 1 0 0 0 0 1
0 0 0 0 0 0 0 0
0 0 0 0 0 0 1 0
### Graph SCC
Visit on G^T:
[ 0] A: dist_time= 1 endp_time=12
[ 1] B: dist_time= 3 endp_time= 4
[ 2] C: dist_time= 2 endp_time=11
[ 3] D: dist_time= 6 endp_time= 7
[ 4] E: dist_time= 8 endp_time= 9
[ 5] F: dist_time= 5 endp_time=10
[ 6] G: dist_time=13 endp_time=16
[ 7] H: dist_time=14 endp_time=15
Visit on G:
[ 0] A: dist_time= 5 endp_time=14
[ 1] B: dist_time= 9 endp_time=10
[ 2] C: dist_time= 8 endp_time=11
[ 3] D: dist_time=15 endp_time=16
[ 4] E: dist_time= 6 endp_time=13
[ 5] F: dist_time= 7 endp_time=12
[ 6] G: dist_time= 1 endp_time= 2
[ 7] H: dist_time= 3 endp_time= 4
Number of SCCs: 4
SCC0: [ 6] G
SCC1: [ 7] H
SCC2: [ 0] A [ 1] B [ 2] C [ 4] E [ 5] F
SCC3: [ 3] D
```

Response
ABCEF D G H

Exercise 6 (1.5 points)

Given the following undirected and weighted graph find a minimum spanning tree using Prim algorithm. Start from vertex A.



Indicate the total weight of the final minimum spanning tree. Report one single integer value. No other symbols must be included in the response. This is an example of the response format: 13

Solution

UN-directed matrix ... stated as such.

Weighted matrix ... stated as such.

Vertex list: [0] A [1] B [2] C [3] D [4] E [5] F [6] G [7] H [8] I [9] J

Adjacency Matrix:

```

0 1 1 0 0 0 0 0 0 0 1
1 0 7 7 2 3 0 0 0 2 0
1 7 0 7 0 0 0 0 0 0 0
0 7 7 0 7 0 7 0 0 0 0
0 2 0 7 0 2 0 3 0 1
0 3 0 0 2 0 0 0 0 0 5
0 0 0 7 0 0 0 0 0 0 1
0 0 0 0 3 0 0 0 0 0 0
0 2 0 0 0 0 0 0 0 0 0
1 0 0 0 1 5 1 0 0 0 0

```

Graph MST (Kruskal and Prim)

Kruskal

Sorted array of edges: src dst weight:

```

A B 1
A C 1
A J 1
E J 1
G J 1
B E 2
B I 2
E F 2
E H 3
E J 3
F J 5
B C 7
B D 7
C D 7
D E 7
D G 7

```

MST (list of edges):

```

Edge [ 0] A - [ 0] B ==> weight = 1
Edge [ 0] A - [ 0] C ==> weight = 1
Edge [ 0] A - [ 0] J ==> weight = 1
Edge [ 4] E - [ 4] J ==> weight = 1
Edge [ 6] G - [ 6] J ==> weight = 1
Edge [ 1] B - [ 1] I ==> weight = 2
Edge [ 4] E - [ 4] F ==> weight = 2
Edge [ 4] E - [ 4] H ==> weight = 3
Edge [ 1] B - [ 1] D ==> weight = 7
Total tree weight = 19

```

Prim

MST (list of edges):

```

Edge [ 0] A - [ 1] B ==> weight = 1
Edge [ 0] A - [ 2] C ==> weight = 1
Edge [ 0] A - [ 9] J ==> weight = 1
Edge [ 9] J - [ 4] E ==> weight = 1
Edge [ 9] J - [ 6] G ==> weight = 1
Edge [ 4] E - [ 5] F ==> weight = 2
Edge [ 1] B - [ 8] I ==> weight = 2
Edge [ 4] E - [ 7] H ==> weight = 3

```

```
Edge [ 1] B - [ 3] D ==> weight = 7
Total tree weight      = 19
```

Response

19

Exercise 7 (4.0 points)

Describe how to perform a Depth First Search on the graph G and report the pseudo-code of a standard DFS procedure. Moreover, define how to label edges (as Tree, Forward, Backward, and Cross) for directed.

Solution

Please, see the teacher's overhead for a complete analysis of the topic.

Exercise 8 (2.0 points)

Analyze the following program and indicate the exact output it generates. Please, report the exact program output with no extra symbols.

```
int f (int [], int);
int main(void) {
    int i, l, v[32];
    l = f (v, 23);
    for (i=0; i<l; i++)
        printf ("%d", v[i]);
    return (l);
}
int f (int v[], int n) {
    int r, l;
    l = 0;
    while (n!=0) {
        r = n & 01;
        n = n >> 1;
        v[l++] = r;
    }
    return (l);
}
```

Response

11101

Exercise 9 (2.0 points)

The following function prints a powerset.

```
int powerset (int *val, int *sol, int n){
    int count, k;
    printf ("{ } \n");
    count = 0;
    for (k=1; k<=n; k++){ // LINE 1
        count += powerset_r (val, sol, n, k, 0, 0);
    }
    return count;
}
int powerset_r (int *val, int *sol, int n, int k, int start, int pos) {
    int count = 0, i;
    if (pos >= k){
        printf("{ ");
        for (i=0; i<k; i++)
            printf("%d ", sol[i]);
        printf("} \n");
        return 1;
    }
    for (i=start; i<n; i++){
        sol[pos] = val[i]; // LINE 2
        count += powerset_2_r (val, sol, n, k, i, pos+1); // LINE 3
    }
    return count;
}
```

Indicates which ones of the following statements are correct. Note that incorrect answers imply a penalty in the final score.

1. Function `powerset_r` implements the simple combinations.
2. In LINE 3, the value `i` must be substituted by `i+1`.
3. Function `powerset_r` implements the arrangements with repetition principle.
4. Function `powerset_r` implements the combination with repetition principle.
5. In LINE 1, `k` must vary from 0 to `n`.
6. In LINE 3, the value `pos+1` must be substituted by `pos`.
7. LINE 2 must be `sol[i]=val[pos]`

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.

Response

FALSE. FALSE. FALSE. TRUE. TRUE. FALSE. FALSE.

Exercise 10 (2.0 points)

Analyze the following program and indicate the exact output it generates. Please, report the precise program output with no extra symbols.

```
int f (int, int);
int f (int n, int m) {
    int l1=0, l2=0;
    if (n<=0 && m<=0) {
        return 0;
    }
    if (n>0) {
        l1 = f (n-1, m);
    } else {
        if (m>0)
            l2 = f (n, m-1);
    }
    return (l1+l2+1);
}
int main () {
    int rv;
    rv = f(5,6);
    fprintf (stdout, "%d", rv);
    return 1;
}
```

Response

11

Exercise 11 (3.0 points)

A square matrix, with `r` rows and `c` columns, stores integer values.

Write the function

```
int max_diff (int **mat, int r, int c);
```

which receives the matrix `mat` and its size (`r`, `c`) and it displays the indices (row and column) of the element for which the difference between the maximum and minimum elements belonging to the diagonal and the inverse-diagonal is maximum.

For example, with $r = 4$, $c = 5$, and the matrix represented on the left-hand side of the following picture, we have to print the values (2,3) as represented on the right-hand side picture. For this value all elements of the diagonal and inverse diagonal are represented with a gray background. Among all these values the program must select the maximum and the minimum value, i.e., 5 and -2 , and compute the difference, i.e., $(5 - (-2)) = 7$. Write the entire program

	0	1	2	3	4
0	0	-2	1	0	0
1	0	0	1	0	2
2	0	0	0	1	1
3	1	0	5	0	0

	0	1	2	3	4
0	0	-2	1	0	0
1	0	0	1	0	2
2	0	0	0	1	1
3	1	0	5	0	0

using standard C libraries but implement all required personal libraries. Modularize the program adequately, and report a brief description of the data structure and the logic adopted in plain English. Unclear or awkward programs, complex or impossible to understand, will be penalized in terms of the final evaluation.

Solution

To be done.

Exercise 12 (5.0 points)

An n -ary tree stores element with integer keys. We define as n the number of nodes at the same depth d in the tree (please, remind that in a tree the root has depth zero).

Write the function

```
int display_crowded_depth (node_t *root);
```

that receives the pointer to the root of such a tree and returns the depth value d for which n is maximum. The function must also display (in any order) all node keys at depth d . The candidate must define the `node_t` structure. For example, if the variable `root` points to the tree represented in the following picture the number of nodes n at depth 2 equals 5; it is the maximum number of nodes with the same depth. Thus, the function must display the following values (in any order) and returns 2. 1 5 12 7 4

Write the entire program using standard C libraries but implement all required personal libraries. Modularize the program adequately, and report a brief description of the data structure and the logic adopted in plain English. Unclear or awkward programs, complex or impossible to understand, will be penalized in terms of the final evaluation.

Solution

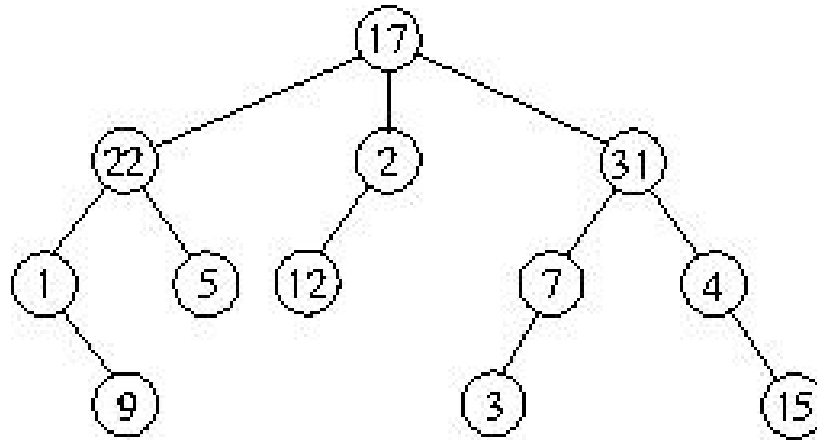
To be done.

Exercise 13 (9.0 points)

Write a function to partition an array of integer values in a set of blocks satisfying some specific properties.

The prototype of the function is the following one:

```
void partition (int *vet, int n, int k);
```



The function receives:

- An array on integer values vet.
- The size of the array n.
- An integer value k (smaller or equal to n).

and it partitions the n values stored in vet in k blocks (i.e., subsets) such that:

1. The number of values included in each block varies by at most one.
2. If S_i is the sum of the values contained in each block, the variance of the S_i values must be minimum (among all possible partitions of the values).

Please, remind that given k values S_i their average and variance can be computed as:

$$A(S_i) = (S_1 + S_2 + \dots + S_k)/k$$

$$V(S_i) = (\sum (S_i - \text{mean}(S_i))^2)/(k - 1)$$

For example, let us suppose that $\text{vet} = 1, 2, 3, 4, 5$, $n = 5$, and $k = 2$. We have to generate all partitions of vet with $k=2$ blocks. A few of these partitions are the following: $\{(1), (2, 3, 4, 5)\}$, $\{(1, 2), (3, 4, 5)\}$, $\{(3, 5), (1, 2, 4)\}$, etc. For the first one, i.e., $\{(1), (2, 3, 4, 5)\}$, the first block has one element and the second block has four elements; thus, the partition does not satisfy constraint number 1 and must be discarded. For the last one, i.e., $\{(3, 5), (1, 2, 4)\}$, $S_1 = (3 + 5) = 8$ and $S_2 = (1 + 2 + 4) = 7$, thus:

$$A(S_i) = (8 + 7)/2 = 7.5$$

$$V(S_i) = ((8 - 7.5)^2 + (7 - 7.5)^2)/(k - 1) = (0.25 + 0.25)/1 = 0.5$$

The program must display the partition of the values in vet for which the value $V(S_i)$ is minimum.

Write the required function program using standard C libraries but implement all required personal libraries. Modularize the program adequately, and report a brief description of the data structure and the logic adopted in plain English. Unclear or awkward programs, complex or impossible to understand, will be penalized in terms of the final evaluation.

Solution

To be done.